Solution:

Take conveniently PQ as x-axis, PA as y-axis and P is the origin (instead of conventional origin O). Therefore, the coordinates are $P(0, 0)$, $A(0, 3)$ and $B(6, 5)$

If the image of A about the x-axis is $\overline{\text{A}}$, then $\overline{\text{A}}$ is $(0, -3)$.

The required R is the point of intersection of the line $\overline{\text{A}}\text{B}$ and x-axis.

AR and BR are the path of the cable (road)

The shortest length of the cable is $AR + BR = BR + R\overline{\text{A}} = B\overline{\text{A}} = \sqrt{(6-0)^2 + (5+3)^2} = 10$ km

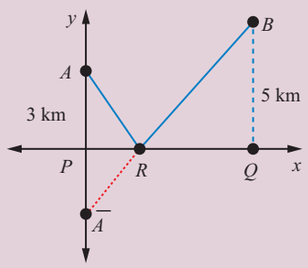


Figure 6.45

Equation of the line $\overline{A}B$ is $y - (-3) = \dfrac{5-(-3)}{6-0}(x-0)$

$$4x - 3y = 9$$

When $y = 0, R$ is $\left(\dfrac{9}{4}, 0\right)$

That is the substation should be located at a distance of 2.25 km from P.

The equation of AR is $4x + 3y = 9$

The equations of the cable lines (roads) of RA and RB are $4x - 3y = 9$ and $4x + 3y = 9$.

Example 6.31 A car rental firm has charges ₹ 25 with 1.8 free kilometers, and ₹ 12 for every additional kilometer. Find the equation relating the cost y to the number of kilometers x. Also find the cost to travel 15 kilometers

Solution:

Given that up to 1.8 kilometers the fixed rent is ₹ 25.

The corresponding equation is

$$y = 25, \quad 0 \le x \le 1.8 \tag{6.26}$$

Also ₹ 12 for every additional kilometer after 1.8 kilometers

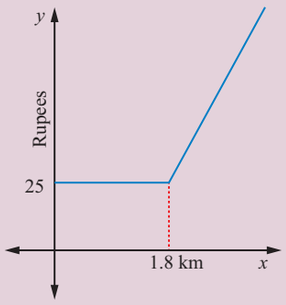


Figure 6.46

The corresponding equation is

$$y = 25 + 12\,(x - 1.8)\,, \quad x > 1.8 \tag{6.27}$$

The combined equation of (6.26) and (6.27), we get

$$y = \begin{cases} 25, & 0 \le x \le 1.8 \\ 25 + 12(x - 1.8), & x > 1.8 \end{cases} \tag{6.28}$$

When $x = 15$, from (6.27), we get cost to travel 15 kilometers is ₹ 183.40.